

Bachelor AI - MLOps

11 PROCESSING AND PREDICTION MODES, FEATURES

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Three fundamental ways data is processed

- Request-Response
- Batch Processing
- Stream Processing

- stateless
- synchronous
- easily scalable

- stateful
- bounded
- parallelizable

- stateful
- unbounded
- complex

- Batching and streaming are made for stateful, high-volume processing
- Request-Response
 - rather used for user-facing apps
 - mostly not suitable for data processing, as it is normally stateless
 - can however also load and use results from batch processing

- **Features:** The inputs of an ML model
- **Feature Engineering:** Calculation of features from raw data
- **Unstructured Data:** (rather less feature engineering)
- **Structured Data:** (rather more feature engineering)

- Batch Features
- Streaming Features
 - Real-Time (RT) Features
 - Near-Real-Time (NRT) Features

Feature Types 2/2

Feature Type	Calculation	Compute Engine	Latency	Advantages	Disadvantages
		(Ex.)			
Batch	Pre-calculated in batch process	Spark	Hours to days	simple, parallelizable	no current features
RT	At the time of prediction	Python, SQL	<1s	simple, current features	doesn't scale
NRT	Pre-calculated in streaming process	Flink, Quix	Secs to mins	Current features, scalable	difficult to implement

- Only requires a (often already existing) batch processing infrastructure
- Advantages
 - Low complexity
 - Parallelizable
- Disadvantages
 - Limited actuality
 - Possibly unnecessary calculations

- Calculated directly within each request
- Advantage
 - No time delay, 100% current
 - Easy to implement
- Disadvantage
 - Calculation time directly increases latency, therefore scales poorly

- Calculated asynchronously (in contrast to RT)
- Latency is only increased by lookup time
- Difference to Batch:
 - Tends to be calculated more frequently
 - Calculated using stream processing

We distinguish different levels

- 1 Offline Prediction (Batch Prediction)
- 2 Online Prediction
 - 1 exclusively with Batch Features
 - 2 with Streaming and Batch Features
- 3 Continual Learning

Offline Prediction (Batch Prediction)

- Calculations are performed periodically and in advance (before use)
- Some typical use cases:
 - Lead Scoring
 - Demand Forecasting
 - Inventory Management
- Prerequisites are a batch infrastructure
- Model Registry would be a possible expansion stage

- Online Prediction: When the prediction is only made when it is needed
- Only pre-calculated features are used
- When batch features are (temporally) *accurate enough*
- In addition to batch infrastructure, a fast key-value store is needed

- RT, NRT and Batch Features can be combined
- Prerequisite is a stream processing infrastructure

Online-Prediction: Aggregated Features vs. RT Features

- Some features are calculated by aggregating over many rows
 - Average purchase amount per user
 - Median income by zip
- Other features are calculated for just one row
 - A ratio between two existing columns
 - Binning (temperature, age)
 - Time-related (days since last login)
 - Counts (number of sentences)

This difference is fundamental

- The former are calculated in advance and are therefore Batch or NRT features
- The latter can (and sometimes must) be calculated as RT feature

- Almost always the environment and thus input data for the model change over time
- Then models must be retrained periodically
- This requires not only input data (X) but also the target (y)
- Sometimes the target arises automatically in the process (so-called Natural Labels)
- Often it must be elaborated manually

- Manual stateless retraining
- Automated stateless retraining
- Automated stateful training
- Continual Learning

- Retraining when the need arises
- Labelling of new training data is done manually
- Compilation of training data manual
- With old and new data or only with new data
- Stateless: Model is retrained from scratch

- Possible with Natural Labels
- Less error-prone
- Complex when input data and labels arrive with significant time delay

- Also called *incremental learning*
- Continue to train existing models with new current data
- Requires *Model Lineage*

- Complete automation
- If performance decreases too much, retrain automatically
- Key questions:
 - When should retraining occur
 - Is the new model actually better